

The EM (Expectation Maximization) Algorithm

- ❑ The k-means algorithm has two steps at each iteration
 - ❑ Expectation Step (E-step): Given the current cluster centers, each object is assigned to the cluster whose center is closest to the object: An object is expected to belong to the closest cluster
 - ❑ Maximization Step (M-step): Given the cluster assignment, for each cluster, the algorithm adjusts the center so that the sum of distance from the objects assigned to this cluster and the new center is minimized
- ❑ The (EM) algorithm: A framework to approach maximum likelihood or maximum a posteriori estimates of parameters in statistical models
 - ❑ E-step assigns objects to clusters according to the current fuzzy clustering or parameters of probabilistic clusters
 - ❑ M-step finds the new clustering or parameters that minimize the sum of squared error (SSE) or the expected likelihood

Fuzzy Clustering Using the EM Algorithm: Example

- Initially, let $c_1 = a$ and $c_2 = b$
- The first E-step: assign o to c_1 with membership weight

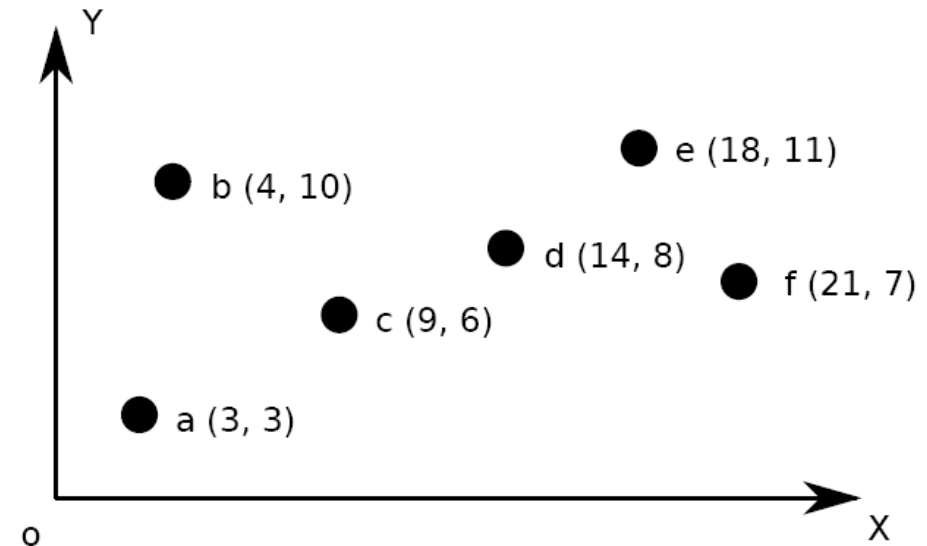
$$\frac{\frac{1}{\text{dist}(o, c_1)^2}}{\frac{1}{\text{dist}(o, c_1)^2} + \frac{1}{\text{dist}(o, c_2)^2}} = \frac{\text{dist}(o, c_2)^2}{\text{dist}(o, c_1)^2 + \text{dist}(o, c_2)^2}$$

$$w_{c,c_1} = \frac{41}{45+41} = 0.48$$

$$w_{d,c_1} = \frac{104}{146+104} = 0.416$$

$$w_{e,c_1} = \frac{197}{289+197} = 0.405$$

$$w_{f,c_1} = \frac{298}{340+298} = 0.467$$



Fuzzy Clustering Using the EM Algorithm: Example

- The first M-step: recalculate the centroids according to the partition matrix, minimizing the sum of squared error (SSE)

$$c_j = \frac{\sum_{\text{each point } o} w_{o,c_j}^2 o}{\sum_{\text{each point } o} w_{o,c_j}^2}$$

- $c_1 = (8.47, 5.12)$
- Iteratively calculate this until the cluster centers converge or the change is small enough

Iteration	E-step	M-step
1	$M^T = \begin{bmatrix} 1 & 0 & 0.48 & 0.42 & 0.41 & 0.47 \\ 0 & 1 & 0.52 & 0.58 & 0.59 & 0.53 \end{bmatrix}$	$c_1 = (8.47, 5.12),$ $c_2 = (10.42, 8.99)$
2	$M^T = \begin{bmatrix} 0.73 & 0.49 & 0.91 & 0.26 & 0.33 & 0.42 \\ 0.27 & 0.51 & 0.09 & 0.74 & 0.67 & 0.58 \end{bmatrix}$	$c_1 = (8.51, 6.11),$ $c_2 = (14.42, 8.69)$
3	$M^T = \begin{bmatrix} 0.80 & 0.76 & 0.99 & 0.02 & 0.14 & 0.23 \\ 0.20 & 0.24 & 0.01 & 0.98 & 0.86 & 0.77 \end{bmatrix}$	$c_1 = (6.40, 6.24),$ $c_2 = (16.55, 8.64)$